

1. Executive Architecture Overview

This High-Level Design (HLD) specifies the software architecture for the Engine Management System (EMS). The system complies with AUTOSAR Classic Platform Release 4.4.0. It defines the software components (SWCs), their interfaces, runnable entities, and inter-component signal mappings.

Primary components include EngineControl_SWC, ThrottlePosition_SWC, and DiagnosticManager_SWC.

2. Software Component Specifications

2.1 EngineControl_SWC

EngineControl_SWC calculates fuel injection timing and ignition angle. It receives throttle position input signals and outputs actuator control commands to ECU drivers.

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ThrottlePosition_SWC reads raw sensor voltages from ADC and produces ThrottlePosition_Signal.

3. Interface and Signal Mapping Matrix

Signal Name	Source SWC	Target SWC	Data Type	Interface Type
ThrottlePosition_Signal	ThrottlePosition_SWC	EngineControl_SWC	float32	Sender-Receiver
EngineSpeed_Signal	SensorAcquisition_SWC	EngineControl_SWC	uint16	Sender-Receiver
FuelInjectionCmd_Signal	EngineControl_SWC	ActuatorDriver_SWC	float32	Sender-Receiver
OrphanedSensor_Signal	TempSensor_SWC	UNMAPPED	uint8	Sender-Receiver
DiagStatus_Signal	DiagnosticManager_SWC	EngineControl_SWC	boolean	Client-Server